

PROJECT PROPOSAL

Title

Heuristic Approach to Storage Resource Allocation for Cost Reduction and Service Level Agreement-Aware Availability

1. Introduction

The rapid adoption of cloud computing has significantly increased the demand for efficient storage resource management. Organizations rely on cloud storage services to provide reliable data availability while maintaining operational costs within acceptable limits. However, improper allocation of storage resources often leads to over-provisioning, under-utilization, increased operational expenses, and potential violations of Service Level Agreements (SLAs).

Traditional storage allocation techniques may not adequately balance cost optimization and availability requirements. Heuristic approaches offer practical solutions by generating near-optimal allocation decisions within reasonable computational time. This project proposes the development of a heuristic-based storage resource allocation system that minimizes storage costs while ensuring SLA-aware availability requirements are met.

To demonstrate and evaluate the effectiveness of the proposed approach, a software platform will be developed using Streamlit, providing visualization, monitoring, recommendation, and reporting functionalities.

2. Problem Statement

Cloud service providers and organizations frequently face challenges in allocating storage resources efficiently. Excessive resource allocation results in unnecessary costs, while insufficient allocation can lead to SLA violations and reduced service availability.

There is a need for an intelligent allocation mechanism capable of balancing storage cost and availability requirements while providing administrators with actionable recommendations and real-time visibility into resource utilization.

3. Aim

To design and implement a heuristic-based storage resource allocation system that reduces storage costs while maintaining SLA-aware availability requirements.

4. Objectives

1. To study existing cloud storage allocation strategies and SLA management techniques.
2. To design a heuristic algorithm for storage resource allocation.
3. To develop a software platform for storage monitoring and optimization.
4. To implement cost estimation and SLA compliance evaluation mechanisms.
5. To generate storage allocation recommendations based on workload requirements.

6. To evaluate the performance of the proposed heuristic approach against baseline allocation methods.

5. Scope of the Project

The project will focus on:

- Cloud storage resource allocation.
- Cost optimization strategies.
- SLA-aware availability management.
- Workload-based allocation recommendations.
- Storage utilization monitoring.
- Reporting and analytics.

The project will not focus on:

- Physical hardware management.
- Network optimization.
- Real-time production cloud deployment.

6. Methodology

Phase 1: Requirement Analysis

- Review literature on storage resource allocation.
- Study SLA metrics and cloud storage pricing models.
- Identify key optimization parameters.

Phase 2: System Design

Design the architecture consisting of:

- User Interface Layer
- Allocation Engine
- Cost Analysis Module
- SLA Evaluation Module
- Reporting Module
- Database Layer

Phase 3: Heuristic Algorithm Development

Inputs:

- Storage demand
- Availability requirements
- Budget constraints
- Access frequency

Outputs:

- Recommended storage allocation
- Estimated cost

- Expected availability level

Phase 4: Software Development

Develop a Streamlit-based application containing:

Dashboard

- Total storage allocation
- Resource utilization
- Cost overview
- SLA compliance metrics

Allocation Module

Users provide:

- Required storage size
- Expected workload
- Availability requirement

System generates:

- Allocation recommendation
- Cost estimate
- Availability prediction

Monitoring Module

- Storage growth trends
- Utilization charts
- Cost analysis charts

Reporting Module

- SLA reports
- Allocation reports
- Cost optimization reports

Phase 5: Testing and Evaluation

Evaluate:

- Cost reduction percentage
- SLA compliance rate
- Resource utilization efficiency
- Algorithm execution time

7. Proposed Technologies

Frontend:

- Streamlit

Backend:

- Python

Database:

- SQLite or PostgreSQL

Libraries:

- Pandas
- NumPy
- Plotly
- Matplotlib
- Scikit-learn (optional)

Development Tools:

- VS Code
- GitHub

8. Expected Outcomes

The project is expected to:

- Reduce storage allocation costs.
- Improve storage utilization efficiency.
- Maintain SLA-aware availability requirements.
- Provide intelligent allocation recommendations.
- Offer administrators a user-friendly monitoring dashboard.

9. Significance of the Study

This research contributes to cloud resource optimization by demonstrating how heuristic techniques can improve storage allocation decisions. The developed software platform provides a practical tool for evaluating allocation strategies and supporting decision-making in cloud environments.

10. Deliverables

- Heuristic allocation algorithm.
- Streamlit-based web application.
- Storage optimization dashboard.
- Cost estimation module.
- SLA monitoring module.
- Final project report.

- User documentation.